

Mohamed H. Mousa, PhD

Neuroengineer & Computational Neuroscientist

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RESEARCH SUMMARY

Neuroengineer and computational neuroscientist with 8+ years of experience at the intersection of clinical research, neural signal acquisition, and biophysical modeling. Builds Python/MATLAB/NEURON/SpikeInterface pipelines to collect and analyze high-density human (HD-EMG) and animal (in-vivo microelectrode array) electrophysiological data, and has independently led an IRB-approved human-subjects clinical study. Author of 16 peer-reviewed publications, including a first-author 2026 Journal of Neuroscience paper, spanning motoneuron biophysics, spinal circuit dynamics, electrical-stimulation modeling, and neural decoding.

EDUCATION

PhD, Engineering (Computational Neuroscience)

Dec 2022

Wright State University

- Advisor: Sherif Elbasiouny, PhD. Dissertation: *Reverse Engineering Spinal Motoneuron Properties in Slice and Whole Cord Preparations Using Computational Modeling*.

MSc, Biomedical Engineering

May 2017

Cairo University

- Advisors: Ahmed Hesham and Sherif Elbasiouny. Thesis: *Computer Modeling of Calcium & Potassium Channels Dendritic Distributions in Spinal Motoneurons*.

BSc, Systems & Biomedical Engineering

May 2013

Cairo University

- GPA 3.94, ranked 1st of 105 students. Graduation project: embedded-based computer-aided diagnosis system for mammography using ML (98.6% accuracy).

ACADEMIC & PROFESSIONAL APPOINTMENTS

Temple University

Oct 2024 – Present

Postdoctoral Fellow Research Associate Philadelphia, PA

- Advisor: Christopher K. Thompson, PhD; collaborates with Michel A. Lemay.
- Built end-to-end pipelines to process, decompose, and analyze 128-channel electrophysiological data using MATLAB, Python, R, and SpikeInterface.
- Conducted high-density electrode recordings from human subjects (HD-EMG) and performs in vivo electrophysiology in a cat model to record high-density signals, across multiple experimental conditions.
- Developed Python-based analysis pipelines to quantify functional connectivity and non-linear dynamics in spinal interneuron networks using graph-theoretic and statistical methods.
- Developing and optimizing 3D biophysical computational models of spinal motoneurons and interneurons, simulating discharge patterns and post-spinal-cord-injury connectivity changes.

Wright State University

Jan 2023 – Oct 2024

Postdoctoral Fellow Dayton, OH

- Led an IRB-approved clinical study to develop a diagnostic biomarker for ALS: enrolled, consented, and performed EMG recordings on 20+ human participants; managed database design and coordination.
- Analyzed and reverse-engineered electrophysiological recordings from human and animal models to identify disease signatures.
- Developed predictive biophysical computational models, using NEURON and MATLAB, to classify ALS-related neural changes.
- Invited to submit a full application for a competitive postdoctoral fellowship (Milton Safenowitz, The ALS Association) after LOI approval; contributed to grant-writing.

Wright State University

Jun 2017 – Dec 2022

PhD Candidate Dayton, OH

- Developed 3D high-fidelity computational models of motoneurons to reverse-engineer experimental electrophysiological data from slice and whole-cord preparations.
- Modeled electrical/extracellular field potentials and characterized how stimulation parameters alter motoneuron excitability.
- Automated feature extraction from electrophysiological datasets using Python and MATLAB, reducing analysis time by 60×.
- Developed mathematical models of ionic channels to improve computational-model accuracy and replicate complex in-silico motoneuron behaviors.

Cairo University

Dec 2014 – Jun 2017

Research & Teaching Assistant Cairo, Egypt

- Taught courses in medical equipment (MRI, X-Ray), bio-electronics, data structures & algorithms, and computer vision (see Teaching Experience).
- Supervised undergraduate BCI graduation projects using P300 brain waves.
- Modeled storage and broadband requirements of medical facilities.

Maadi Military Hospital

2013 – 2014

Field Clinical Engineer Cairo, Egypt

- Supported installation, calibration, and maintenance of MRI, CT, and ventilator systems in a live clinical environment, coordinating directly with physicians and hospital engineering staff.

GRANTS & FELLOWSHIPS

Milton Safenowitz Postdoctoral Fellowship

2023 – 2024

The ALS Association, USA

- Project: *Identifying early biomarkers of ALS via human sEMG and computational modeling*. Requested budget: \$150,000. Letter of Intent approved; invited to submit and submitted a full proposal, which was not funded.

HONORS & AWARDS

- SfN Trainee Professional Development Award, Society for Neuroscience — includes complimentary registration for Neuroscience 2026 (Nov 14–18, Washington, D.C.) and a \$1,000 travel award 2026
- Excellence Award, top PhD graduate student in biomedical engineering, Wright State University 2023
- Science Gateways Community Institute (SGCI) Science Ambassador — \$1,200 travel award 2019
- Travel Award, Professional Development Grant, Wright State University 2018

- Best Paper Award, Cairo International Biomedical Engineering Conference (CIBEC) 2016
- Graduated First in Class with Distinction, Cairo University 2013
- Recipient of Annual Excellence Award (top 5%), Cairo University 2009–2012

Scholarships

- Selected Participant, Myology Course, Nationwide Children's Hospital, Columbus, OH (registration and accommodation fully covered) Aug 31–Sep 4, 2026
- NIH BRAIN Initiative Summer Course — *Models and Neurobiology*, University of Missouri-Columbia June 2017
- Research Assistant Scholar, Neuro Engineering, Rehabilitation & Degeneration Laboratory, Wright State University Summer 2015 & 2016
- Undergraduate Research Assistant, Hochschule Zittau/Görlitz, Germany Summer 2012

PEER-REVIEWED PUBLICATIONS

- [1] **Mousa, M.H.**, Zaback, M., Lemay, M.A., & Thompson, C.K. (2026). Activity and Functional Connectivity of Lumbar Spinal Interneurons at Rest: Irregular, Sparse Yet Structured. *Journal of Neuroscience*, published online 27 Aug 2026, e0784262026. DOI: 10.1523/JNEUROSCI.0784-26.2026.
- [2] Abdul Halim, I., Gerber, K.R., Gelford, W.B., Ellington, C.L., **Mousa, M.H.**, Garrett, T.L., & Elbasiouny, S.M. (2026). SK channel upregulation and sex-specific mechanisms drive spinal motoneuron reduced excitability with age. *Frontiers in Aging Neuroscience*.
- [3] Zaback, M., **Mousa, M.H.**, & Mildren, R.L. (2025). Shining light on spinal co-ordination: internal states shape interneuron recruitment of muscle synergies. *The Journal of Physiology*.
- [4] **Mousa, M.H.** & Elbasiouny, S.M. (2025). A Multi-Scale, High-Fidelity Computational Model of the Mouse Triceps Surae Motor Pool. *47th Annual International Conference of the IEEE EMBS (EMBC)*.
- [5] **Mousa, M.H.**, Wages, N.P., & Elbasiouny, S.M. (2024). Onion skin is not a universal firing pattern for spinal motoneurons: simulation study. *Journal of Neurophysiology*, 132(1), 240–258.
- [6] Wages, N.P., **Mousa, M.H.**, Clark, L.A., Tavoian, D., Arnold, W.D., Elbasiouny, S.M., & Clark, B.C. (2024). Reductions in motor unit firing are associated with clinically meaningful leg extensor weakness in older adults. *Calcified Tissue International*, 114(1), 9–23.
- [7] Moustafa, M., **Mousa, M.H.**, Saad, M.S., Basha, T., & Elbasiouny, S.M. (2023). Bifurcation analysis of motoneuronal excitability mechanisms under normal and ALS conditions. *Frontiers in Cellular Neuroscience*, 17, 1093199.
- [8] Gamal, M., **Mousa, M.H.**, Eldawlatly, S., & Elbasiouny, S.M. (2021). In-silico development and assessment of a Kalman filter motor decoder for prosthetic hand control. *Computers in Biology and Medicine*, 132, 104353.
- [9] **Mousa, M.H.** & Elbasiouny, S.M. (2021). Estimating the effects of slicing on the electrophysiological properties of spinal motoneurons under normal and disease conditions. *Journal of Neurophysiology*, 124(4), 1450–67.
- [10] **Mousa, M.H.** & Elbasiouny, S.M. (2020). Dendritic distributions of L-type Ca^{2+} and SK_L channels in spinal motoneurons: A simulation study. *Journal of Neurophysiology*, 124(4), 1285–1307.
- [11] Abdelaal, A.Y., **Mousa, M.**, Gamal, M., Elbasiouny, S.M., & Eldawlatly, S. (2020). A Classification Approach to Recognize the Firing of Spinal Motoneurons in Amyotrophic Lateral Sclerosis. *42nd Annual International Conference of the IEEE EMBS (EMBC)*, Montreal, QC, Canada, pp. 3680–3683.
- [12] Elbasiouny, S.M. & **Mousa, M.** (2020). Morphologically-detailed Cellular and Pool Motoneuron Models. In: *Encyclopedia of Computational Neuroscience* (Jaeger D, Jung R, eds). Springer-Verlag Berlin Heidelberg: Springer.
- [13] Mahrous, A.A., **Mousa, M.H.**, & Elbasiouny, S.M. (2019). The Mechanistic Basis for Successful Spinal Cord

Stimulation to Generate Steady Motor Outputs. *Frontiers in Cellular Neuroscience*, 13, 359–374.

- [14] Gamal, M., **Mousa, M.**, Eldawlatly, S., & Elbasiouny, S.M. (2018). Automated Cell-Type Classification and Death-Detection of Spinal Motoneurons. *2018 9th Cairo International Biomedical Engineering Conference (CIBEC)*, IEEE, pp. 57–60.
- [15] **Mousa, M.H.**, Kandil, A.H., & Elbasiouny, S.M. (2016). Simulation of dendritic L-type Ca channels' warm-up phenomenon in spinal motoneurons. *8th Cairo International Biomedical Engineering Conference (CIBEC)*, Cairo, Egypt, pp. 31–34, IEEE.
- [16] ElBially, M.S., Ahmed, M.S., **AbdelGawwad, M.H.**, Ali, F.A., Botros, F.S., & Kadah, Y.M. (2013). K8. Hand-Held Computer Aided Diagnosis System with Application in Mammography. *2013 30th National Radio Science Conference (NRSC)*, pp. 549–556, IEEE. (Published under Mohamed's prior name, AbdelGawwad, M.H.)

INVITED TALKS & PRESENTATIONS

- Short talk on using HD-EMG to investigate motoneuron firing in humans. Myology Course, Nationwide Children's Hospital, Columbus, OH Aug 31–Sep 4, 2026
- “Activity and Functional Connectivity of Lumbar Spinal Interneurons at Rest: Sparse Yet Structured.” Neural Sciences Research Day, Temple University Lewis Katz School of Medicine, Philadelphia Dec 2025
- “Introduction to Computational Modeling.” Ministry of Communications and Information Technology, NAID's assistive technologies internship program, Egypt — Webinar, *Invited Speaker* 2025
- “Resting State Activity of Lumbar Spinal Interneurons: Keeping the Motoneurons Warm through Distributed and Functionally Coupled Subpopulations.” 14th International Motoneuron Society Meeting, St. John's, Newfoundland, Canada July 2025
- “The role of computational models in experimental research: predictive vs. explanatory power.” Neuroscience Gateway Satellite Workshop, George Washington University, D.C. — *Invited Talk* Nov 2023
- “Understanding the Dichotomy in Motoneuron Firing Patterns: ‘Onion Skin’ vs ‘AHP’.” Wright State University Research Celebration, Dayton, OH Oct 2023
- “Differences between SK2 and SK3 currents in mouse spinal motoneurons: A simulation study.” Society for Neuroscience, Chicago, IL Nov 2019
- “Dendritic distribution of Cav1.3 and SK channels in spinal motoneurons: a simulation study.” The International Motoneuron Meeting, Boulder, CO June 2018
- “A Review of Below-Knee Socket Materials Using Finite Element Analysis.” 14th Annual Dayton Engineering Sciences Symposium Oct 2018

TEACHING EXPERIENCE

Wright State University (NCBP; Biomedical, Industrial, & Human Factors Engineering), teaching assistant:

- From Neurons to Behavior – in Health and Disease (BME 7380/BMS 9910) 2017–2024
- Computational Neuroscience (NEU 4600/PN 6600) 2021–2024
- Design and Analysis of Engineering Experiments (EGR 7050) 2018

Cairo University (Biomedical & Systems Engineering Dept.), teaching assistant:

- Classic Control Systems (BME sophomore, 78 students) 2016
- Medical Equipment – MRI (BME senior, 62 students) 2017
- Medical Equipment – X-Ray (BME junior, 66 students) 2015
- Data Structure & Algorithms using C (BME sophomore, 76 students) 2015
- Computer Vision (BME senior, 66 students) 2016
- Biomedical Electronics (BME junior, 62 students) 2017

- Volunteer instructor, “Programming Under UNIX Shell in C” summer workshop 2016
- Volunteer instructor, Medical Equipment (Radiology) summer workshop 2015

TECHNICAL SKILLS

Domain Expertise: Computational Neuroscience | Biophysical Modeling & Simulation | Signal & Image Processing | Network & Graph Theory | Statistical Analysis | Nonlinear Dynamics | Clinical Management & IRB Protocols | Machine Learning

Programming Languages: Python | MATLAB | R | C/C++ | NEURON (Hoc, NMODL) | SQL | JavaScript

Libraries & Tools: NumPy | SciPy | Pandas | Matplotlib | Seaborn | Plotly | statsmodels | Pingouin | SAS | Spikinterface | PyTorch (developing)

PROFESSIONAL MEMBERSHIPS

- Member, The American Physiological Society 2020–present
- Member, Society for Neuroscience (SfN) 2015–present
- Member, IEEE Engineering in Medicine and Biology Society (EMBS) 2017
- Member, The Syndicate of Engineers (Egypt), Electrical Engineering 2013–2016
- Member, International Association for the Exchange of Students for Technical Experience (IAESTE) 2012

SERVICE & LEADERSHIP

- Program Chair, IEEE EMBS International Student Conference — ISCEgypt 2017
- Chair, IEEE BME Cairo University Chapter 2017
- Instructor, “C Programming Under Linux Shell” summer workshop 2016
- Coordinator, BME Project Development Competition, Cairo University IEEE EMBS Chapter 2015

Peer Review Service — ad hoc reviewer for *Journal of Neurophysiology*, *Journal of Neural Engineering*, *Frontiers in Neural Circuits*, *Frontiers in Computational Neuroscience*, *PLOS Computational Biology*, and *Advances in Physiology Education*.